Lecture 43 : Project Monitoring and Control (Contd.)

Introduction

In the dynamic landscape of project management, effective monitoring and control are paramount to ensuring successful outcomes. This lecture delves into advanced aspects of project oversight, building upon previous discussions on cost monitoring. We introduce the concept of Earned Value Analysis (EVA), a powerful refinement that has gained significant popularity, particularly in complex projects. EVA provides a structured approach to integrate project scope, cost, and schedule performance, offering a comprehensive view of project health.

Originating from the stringent requirements of the USA's Department of Defense, EVA was developed as a robust set of measures to control projects executed by contractors. Its fundamental premise involves assigning a quantifiable value to each task or work package within a project's Work Breakdown Structure (WBS), based on original expenditure forecasts. This method allows project managers to move beyond simply tracking actual costs against planned costs, enabling a more nuanced understanding of the value earned for the work performed. We will explore the core concepts of EVA, various techniques for assigning earned value, and the crucial role of a baseline budget in establishing a foundation for effective project monitoring.

Core Concepts

Term	Definition	Significance
`Earned Value Analysis (EVA)`	A project management methodology that integrates project scope, cost, and schedule to assess project performance and progress.	Provides a comprehensive view of project health by comparing planned work, actual work performed, and actual costs incurred.
`Planned Value (PV)`	The original budgeted cost for a specific task or work item, representing the value of work scheduled to be completed by a given date.	Serves as the baseline for measuring project progress and cost performance; also known as Budgeted Cost of Work Scheduled (BCWS).

Term	Definition	Significance
`Earned Value (EV)`	The total value credited to a project at a particular point in time, representing the budgeted cost of the work actually performed.	Indicates the true progress of the project in terms of value delivered, regardless of actual cost; also known as Budgeted Cost of Work Performed (BCWP).
`Baseline Budget`	A time-phased budget that shows the forecast growth in earned value over the project's duration, based on the initial project plan.	Essential for setting up EVA, as it provides the planned expenditure and earned value accumulation against which actual performance is measured.
`0/100 Technique`	An earned value assignment method where a task is assigned zero value until it is 100% complete, at which point its full planned value is credited.	Preferred for software development due to difficulty in objectively measuring partial completion, ensuring value is only recognized upon full delivery.
`50/50 Technique`	An earned value assignment method where 50% of a task's planned value is credited upon its start, and the remaining 50% upon its completion.	Suitable for projects with clear start/finish points and upfront costs (e.g., construction), but can give a false sense of security in others.
`Milestone Technique`	An earned value assignment method where value is credited based on the achievement of predefined project milestones.	Effective for long-duration activities, breaking them into manageable stages with clear deliverables, such as completing design or testing phases.
`Percentage Complete Technique`	An earned value assignment method where value is credited based on an objective, measurable	Applicable when work progress can be quantitatively measured (e.g., data entry, number of

Term	Definition	Significance
	percentage of work completed.	units produced), providing a precise reflection of progress.

Detailed Analysis

1. Earned Value Analysis (EVA) Origins and Principles

Earned Value Analysis (EVA) emerged from the US Department of Defense as a sophisticated method to control complex projects undertaken by contractors. It acts as a significant refinement to traditional cost monitoring processes by integrating cost, schedule, and scope into a single performance measurement system. The core principle of EVA is to assign a monetary or effort-based value to each task or work package defined in the Work Breakdown Structure (WBS). This assigned value is derived from the original expenditure forecasts or predictions for that specific work item.

Conceptually, EVA views the value assigned to a task as the "agreed price" a contractor would receive upon completing a unit of work. This initial budgeted cost for a particular item or task is known as the `Planned Value (PV)`, or Budgeted Cost of Work Scheduled (BCWS). When a task has not yet begun, its earned value is typically zero. Upon its completion, the project is credited with the original planned value for that task. The cumulative total of these planned values for all work completed up to a specific point in time is termed the `Earned Value (EV)`, or Budgeted Cost of Work Performed (BCWP). Earned Value can be expressed in various units, such as monetary values (e.g., rupees), staff time (e.g., person-months or person-hours), or as a percentage of the Planned Value.

2. Techniques for Assigning Earned Value

When tasks are in progress—started but not yet completed—a consistent method is required to assign earned value. Several techniques are commonly employed, each suited to different project contexts:

- **0/100 Technique**: In this method, a task is assigned an earned value of zero until it is 100% complete. Only upon full completion is the entire budgeted value credited to the project. This technique is often preferred in software development projects where partial completion can be difficult to objectively measure and may not represent tangible value. It prevents overvaluing incomplete work.

- **50/50 Technique**: This technique assigns 50% of a task's planned value as soon as the task begins, and the remaining 50% upon its completion. It aligns well with contractual arrangements where an initial payment is made to a contractor to cover startup costs (e.g., raw materials, labor mobilization in building construction). However, for software projects, it can create a "false sense of security" by crediting significant value for merely starting an activity, potentially overstating early progress.
- **75/25 Technique**: Similar to the 50/50 technique, this method assigns 75% of the value upon the task's commencement (or delivery of a major component) and the remaining 25% upon its successful completion or installation. This is particularly useful for projects involving the procurement of large equipment, such as generators or servers, where a substantial payment is made upon delivery, with the final payment contingent on successful installation and testing.
- **Milestone Technique**: This approach assigns earned value based on the achievement of specific, predefined milestones within a task or project. For example, in software development, milestones could include the successful completion of requirement analysis (e.g., SRS prepared), design (e.g., UML diagrams complete), coding, or testing. Each milestone is assigned a portion of the total task value, which is credited only when that milestone is reached. For long-duration activities, it's often beneficial to break them into smaller tasks with distinct milestones.
- **Percentage Complete Technique**: This technique is applied when the amount of work completed can be objectively and quantitatively measured. The earned value is then assigned proportionally to the percentage of work completed. For instance, if an information system implementation requires manually typing 10,000 data records into a database, and after one month, 1,000 records have been entered, then 10% of the task's value can be credited. This method provides a precise reflection of progress for tasks with clear, measurable units of work.

3. Baseline Budget Creation

The first crucial step in setting up Earned Value Analysis is to create a **`baseline budget`**. This budget is fundamentally based on the project plan and illustrates the forecasted growth in earned value over time. It serves as the planned trajectory against which actual project performance will be measured.

For staff-intensive projects like software development, earned value is often measured in person-hours, person-months, or workdays, rather than just monetary values.

To prepare a baseline budget, especially when a Gantt chart is available, the following steps are typically followed:

1. **List Tasks**: Identify all project tasks from the Gantt chart.
2. **Budgeted Workdays**: Determine the planned duration (in workdays) for each task. For example, "Specify overall system" might take 34 days.
3. **Scheduled Completion**: Calculate the scheduled completion day for each task, considering dependencies. If "Specify overall system" starts on day 0 and takes 34 days, its scheduled completion is day 34. If "Specify module B" starts after "Specify overall system" and takes 15 days, its scheduled completion is 34 + 15 = 49 days.
4. **Cumulative Workdays**: Sum the budgeted workdays sequentially to find the cumulative workdays at the completion of each task.
5. Task 1 (34 days): Cumulative = 34
6. Task 2 (15 days): Cumulative = 34 + 15 = 49
7. Task 3 (15 days): Cumulative = 49 + 15 = 64
8. Task 4 (20 days): Cumulative = 64 + 20 = 84
   This process continues until the total cumulative workdays for all tasks are calculated (e.g., 237 days for the entire project).
9. **Percentage of Cumulative Earned Value**: Calculate the percentage of the total project value represented by the cumulative workdays at each task's completion. This is done by dividing the cumulative workdays by the total project workdays and multiplying by 100.
10. For Task 1 (34 cumulative days out of 237 total): (34 / 237) * 100 = 14.35%
11. For Task 3 (64 cumulative days out of 237 total): (64 / 237) * 100 = 27.00%
12. For Task 4 (84 cumulative days out of 237 total): (84 / 237) * 100 = 35.44%
    This calculation provides the forecasted growth in earned value over time, culminating in 100% upon project completion.

The baseline budget can then be graphically represented, plotting the cumulative earned value percentage against time (elapsed days). If the 0/100 technique is used, the graph will show a flat line at zero earned value until a task's scheduled completion day, at which point it will jump to reflect the credited value. For example, if the first task completes on day 34, the graph will show 0 earned value until day 34, then a sudden increase to 14.35% (or 34 workdays of earned value). This visual representation clearly depicts the planned accumulation of earned value throughout the project lifecycle.

Key Takeaways

- [ ] Earned Value Analysis (EVA) refines cost monitoring by integrating scope, cost, and schedule performance.
- [ ] Planned Value (PV) is the budgeted cost of work scheduled, while Earned Value (EV) is the budgeted cost of work performed.
- [ ] EVA originated in the US Department of Defense to control contractor projects.
- [ ] The 0/100 technique is suitable for software development, crediting value only upon 100% task completion.
- [ ] The 50/50 and 75/25 techniques are useful for projects with clear upfront costs or equipment delivery, but can misrepresent software progress.
- [ ] Milestone and Percentage Complete techniques offer flexible ways to assign earned value based on specific achievements or measurable progress.
- [ ] A baseline budget, derived from the project plan, forecasts the growth of earned value over time and is crucial for EVA setup.
- [ ] Earned value can be measured in monetary terms, staff hours, or workdays, depending on the project type.
- [ ] Creating a baseline budget involves calculating budgeted workdays, scheduled completion, cumulative workdays, and percentage of cumulative earned value from a Gantt chart.
- [ ] The baseline budget graph visually represents the planned accumulation of earned value, often reflecting the chosen earned value assignment technique.

Conclusion

The concepts of Earned Value Analysis and baseline budgeting provide project managers with indispensable tools for strategic project monitoring and control. By moving beyond simple cost tracking, EVA offers a holistic view of project performance, allowing for early detection of deviations from the plan in terms of both schedule and budget. The various techniques for assigning earned value—from the conservative 0/100 method favored in software development to the more progressive 50/50 or milestone-based approaches—ensure that value is recognized appropriately for different types of tasks and project phases.

The meticulous creation of a baseline budget, often derived from detailed Gantt charts, establishes a clear, time-phased plan for earned value accumulation. This baseline acts as the critical benchmark against which actual project progress and

earned value are continuously compared. Understanding and applying these principles empowers project teams to make informed decisions, implement timely corrective actions, and ultimately enhance the likelihood of project success by maintaining tight control over scope, schedule, and cost. The next steps in project monitoring will involve leveraging these PV and EV values to assess project health and forecast future performance.